

Final Report

Women Economic Group

School of Information Systems and Management

ISM 4930: Data Visualization

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Introduction

Innovation-driven economies rely on knowledge, technological advancement, and human capital to sustain long-term growth. Within this framework, inclusivity is not merely a social objective but an economic necessity. Excluding women from participation in innovation-based sectors reduces the effective utilization of available talent, potentially lowering productivity and slowing economic growth.

This project investigates the relationship between innovation investment, economic strength, and female inclusion across countries. Specifically, we address:

RQ1(Main): *“Do innovation-driven economies exhibit higher levels of women’s participation in the labor market and leadership roles?”*

RQ2: *“Does excluding women from participation create efficiency limitations within this type of economy?”*

RQ3: *What is the relationship between women’s inclusion and economic performance across countries?*

Methodology

Data Sources:

To conduct our analysis, we used a combination of macroeconomic and social data. These datasets allow us to examine our area of interest from multiple perspectives including economy, education, innovation through research and development, and inclusion.

- **World Bank – World Development Indicators (WDI)**
<https://databank.worldbank.org/source/world-development-indicators>

This dataset provides internationally comparable indicators on economic performance, innovation, education, and labor market participation across countries.

- **International Monetary Fund (IMF) – Gender Inequality Data**
<https://www.imf.org/en/Data>

This dataset includes the Gender Inequality Index (GII) and the Gender Development Index (GDI), capturing disparities between men and women across countries.

- **UNESCO Institute for Statistics – R&D Data**
<http://data.uis.unesco.org/>

This source provides data on gross domestic expenditure on research and development as a percentage of GDP.

Variables:

Throughout the development of our project, we were continuously evaluating which variables we are going to use in our visualizations. We had a broad set of variables in the very beginning, but we narrowed it down by the most important and relevant ones. The latest added variables were gross labor force participation rate, and both percentage of female and male in the labor force. Here are the following categories of variables:

- ***Economic Indicators***
 - GDP
 - GDP per capita
 - GDP % growth
- ***Innovation Indicators***
 - Research and development expenditure (% of GDP)
 - Patent applications (residents)
 - Scientific and technical journal articles
 - Trademark applications, resident, by count
 - Scientific and technical journal articles
 - Industrial design applications, resident, by count
- ***Education Indicators***
 - Female tertiary education enrollment (%)
- ***Female Inclusion Indicators***
 - Female labor force participation rate (%)
 - Male labor force participation rate (%)
 - Female-to-male labor force participation ratio
 - Female share of employment in senior and middle management (%)
 - Gender Inequality Index
 - Women Business and the Law Index Score (scale 1-100)
 - Proportion of seats held by women in national parliaments (%)

Data Preparation

Data from the different sources were cleaned and integrated using Tableau Prep. The process included:

- Removing redundant variables to focus only on relevant ones
- Pivoting tables
- Removing unnecessary columns
- Merging datasets using country identifiers
- Incorporating gender-disaggregated variables to enable direct comparisons
- Standardizing indicators to ensure cross-country comparability

The final dataset was structured across six linked tables to support the analysis across all dimensions of economy, innovation, education, and female inclusion.

The analysis is structured to evaluate whether innovation-driven economies are also associated with greater female inclusion. The visualizations are organized sequentially to move from a broad macroeconomic perspective toward more specific measures of innovation, human capital development, and gender-based outcomes.

Analysis

Economic Landscape

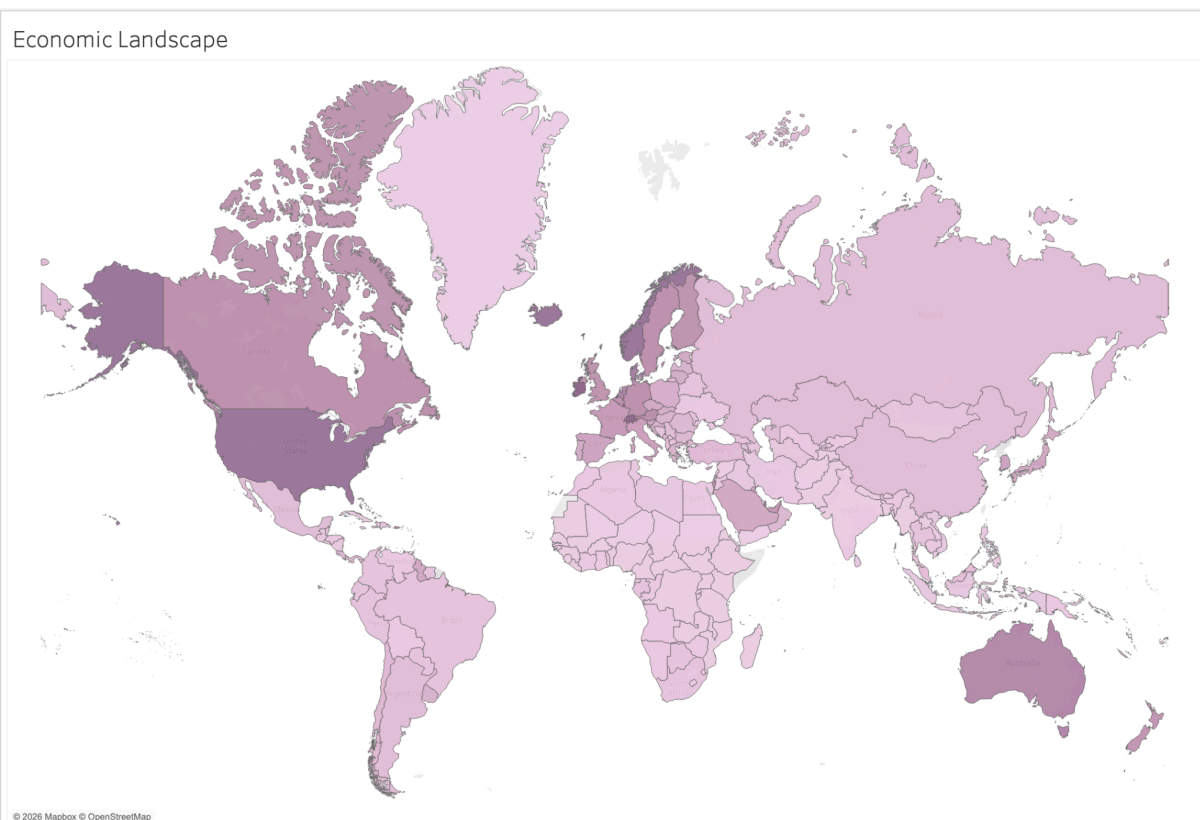


Figure 1. Economic Landscape – GDP per Capita by Country

The global map (Figure 1) reveals a stark geographic divide in economic strength: North America, Western Europe, and Australia/New Zealand dominate in GDP per capita, while Sub-Saharan Africa and South/Southeast Asia occupy the lowest range. This distribution establishes the economic baseline against which innovation investment and gender inclusion are assessed. Notably, high GDP per capita is concentrated in a relatively small cluster of countries, which are also the primary candidates for high innovation investment. We wanted to include this because of the role economic strength has in shaping a country's capacity. Greater capacity indicates the likelihood that countries can invest in innovation while also building inclusive systems.

Innovation

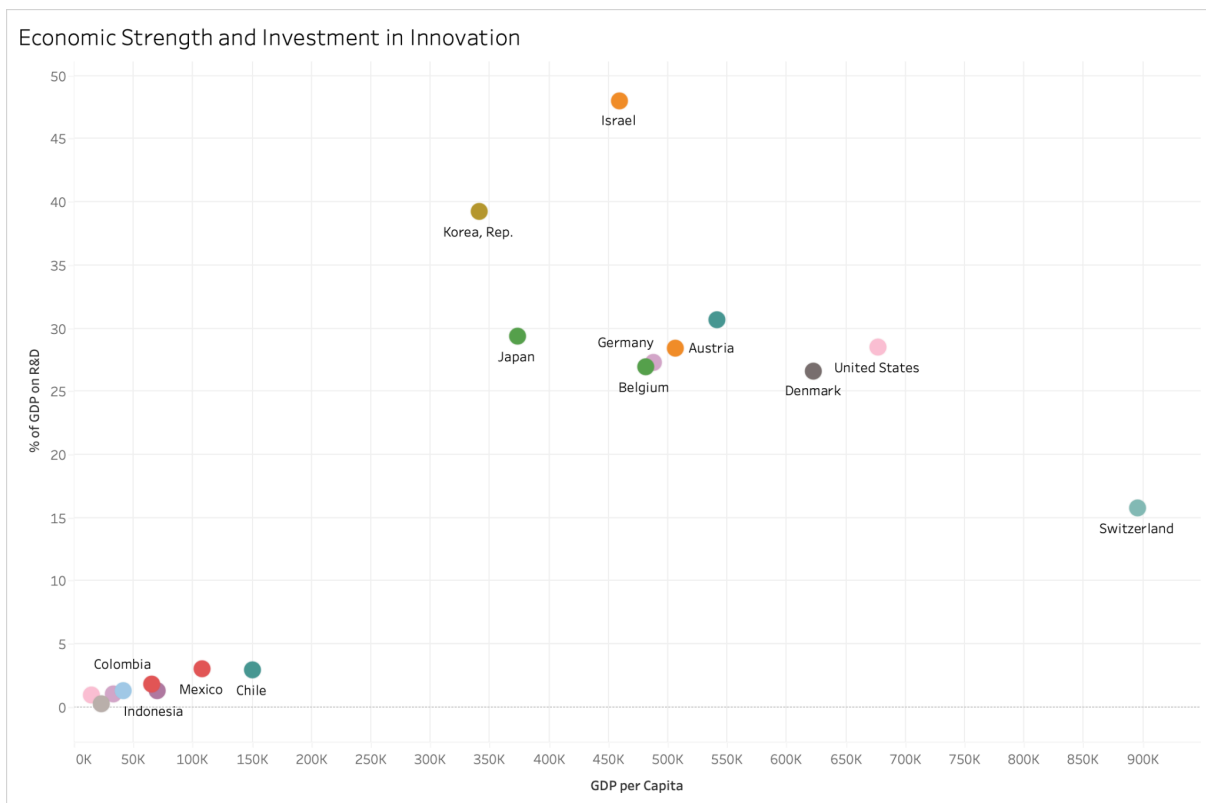


Figure 2. Economic Strength and Investment in Innovation (R&D % of GDP vs GDP per Capita)

The Figure 2 scatter plot reveals that higher levels of investment in innovation does not solely rely on the country's overall economic strength. We can see that Israel and Korea have a larger share of their GDP dedicated to R&D, while others may not prioritize or have the means to invest higher. What we are able to suggest is that the investment reflects a difference in a strategy and having access to resources that can support an innovative system. We could assume that the bottom ten countries have a limited capacity. The takeaway is the difference in how the top and bottom ten are able allocate resources that lead to stronger opportunities.

Innovation Cont.

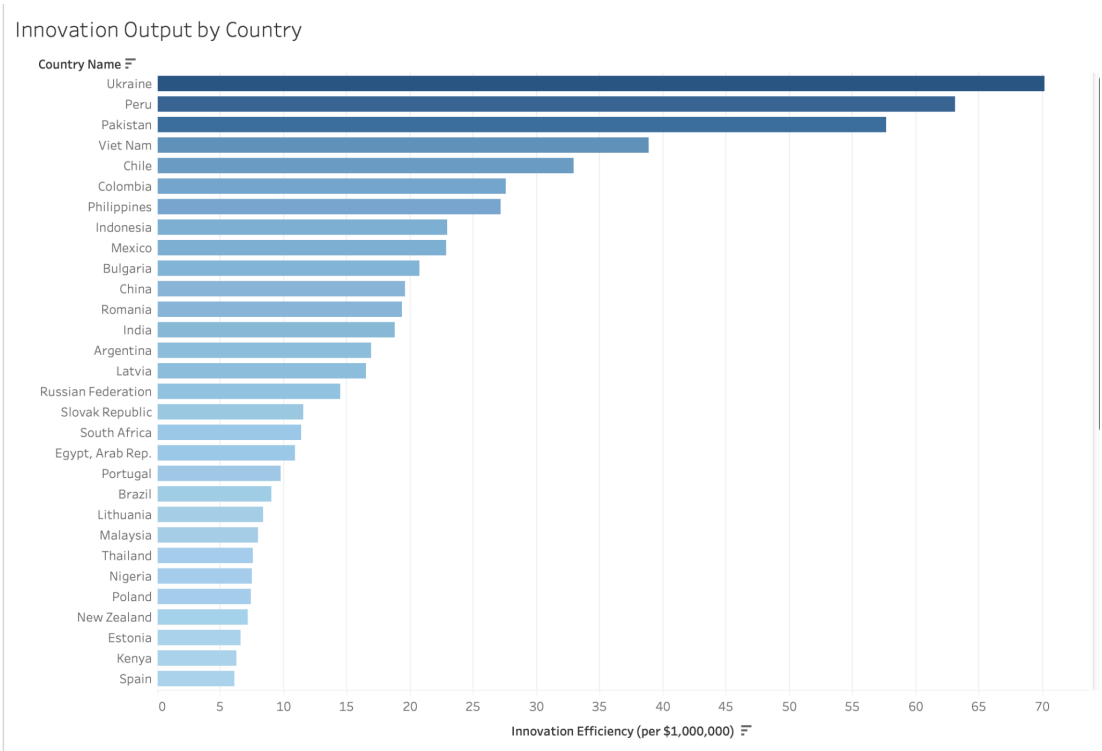


Figure 3. Innovation Output by Country (Innovation Efficiency per \$1,000,000 of R&D)

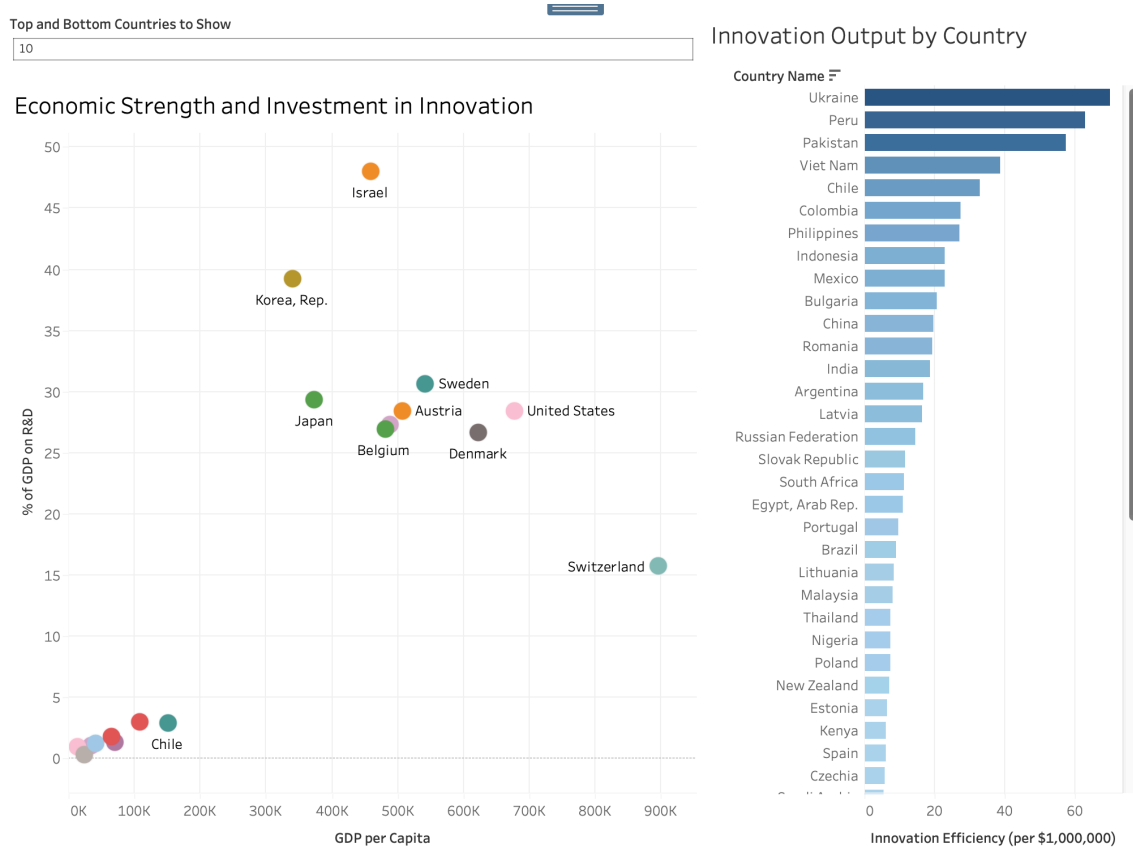


Figure 4. Innovation Dashboard – Economic Strength, Investment, and Output

Figure 3 measures innovation efficiency by comparing total innovation outputs, including patents and trademarks, to a country's GDP and R&D investment, allowing for comparison across countries of different economic sizes. Ukraine leads this metric, followed by Peru and Pakistan, indicating that these countries generate relatively more innovation output per unit of R&D spending. Although, this does not completely mean that they have stronger innovation systems, but instead suggests how their efficiency is a reflection of how well they use their resources.

This measure does not capture the overall impact of innovation, but instead highlights how effectively countries convert their resources to outcomes. Some countries may be able to produce outcomes quicker due to more efficient regulatory processes. Our takeaway is that efficiency speaks to how fast countries can convert inputs to outputs and how that influences available opportunities.

When looking at them together in the Innovation Dashboard (Figure 4) we can see how scale and efficiency of the investment are both needed to analyze how well that innovation system works. Together they work in shaping opportunities for inclusion and participation, but it's up to how the government processes patents, trademarks, etc faster.

Education

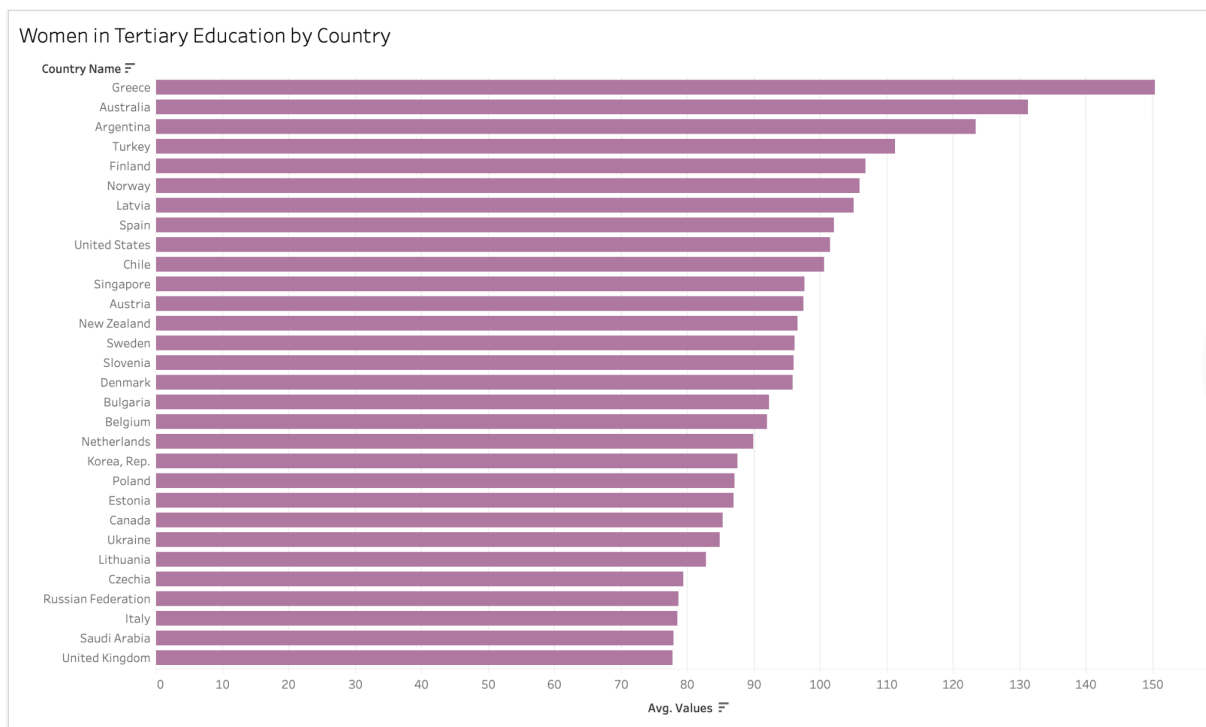


Figure 5. Women in Tertiary Education by Country (Avg. Enrollment Rate)

Figure 5 shows female tertiary education enrollment rates by country. Crucially, values exceeding 100% mean that UNESCO counts every person who is enrolled in higher education, regardless of being out of the college-age group. Several high-innovation economies such as Korea, Sweden, and Denmark display enrollment rates in the 85–95% range, indicating strong but not top-ranked female access to higher education. This tells us that there are well-established education systems that have the means and framework to support broad participation. For example, with Greece ranking so high is a reflection of their structural factors that allows them to have more public access. It breaks down to the difference of each country's structural approach to education. This finding leads us to suggest that innovation leadership does not strictly require the highest female tertiary enrollment, but that a broad baseline of access is another layer of support for participation in the workforce.

Female Inclusion

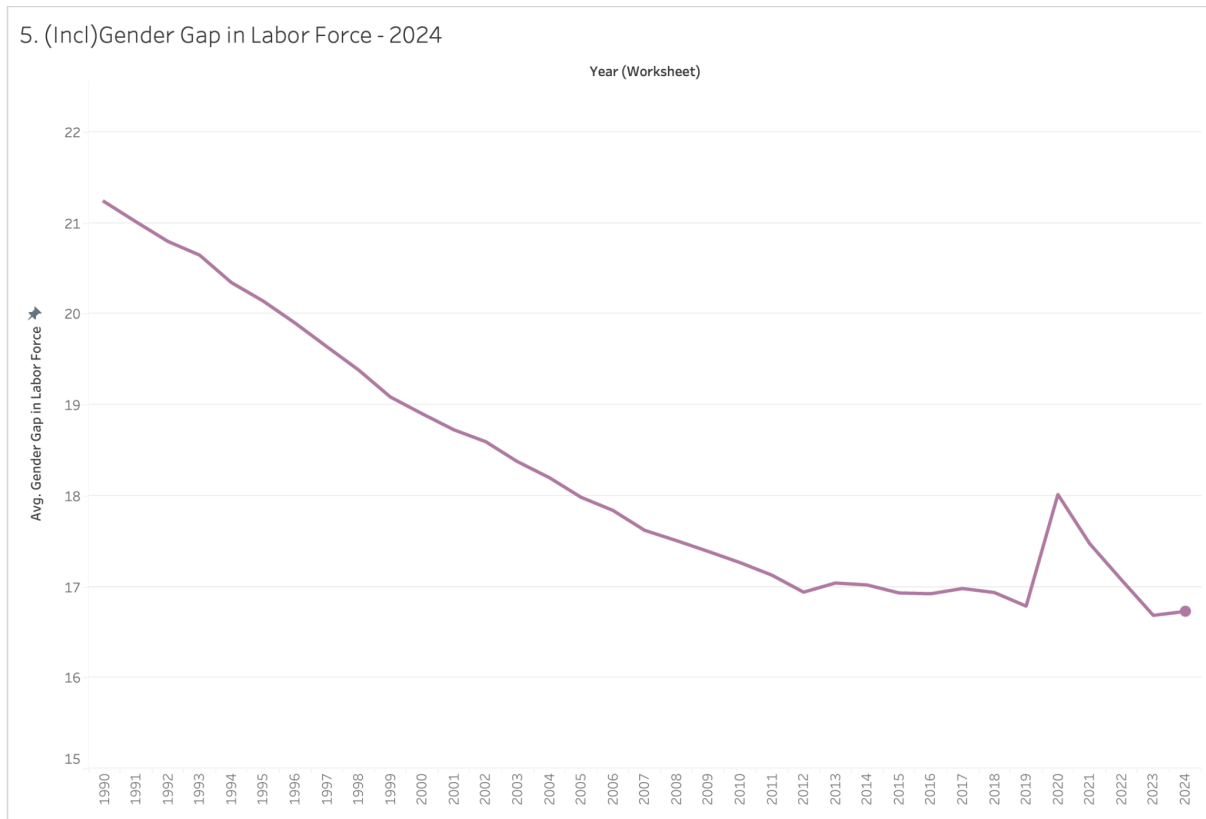


Figure 6. Gender Gap in Labor Force Participation (1990–2024, Animated)

Figure 6 tracks the average global gender gap in labor force participation from 1990 to 2024. The data reveals the gradual decline of the gap which we can see is an ongoing pattern starting around the 21 mark in 1990 to the 17th mark in 2019. The only disruption to the pattern being 2019 once COVID hit, but returned to its pre-pandemic trajectory shortly after. This reflects a continuous improvement regarding workforce opportunities. Although, the persistence of the size gap tells us how structural barriers still continue to limit full participation. These barriers could be a result of unequal access to leadership roles, differences in unpaid care responsibilities and labor market segmentation which may maintain the disparity across countries.

Female Inclusion Cont.

Gender Inequality Index by Country

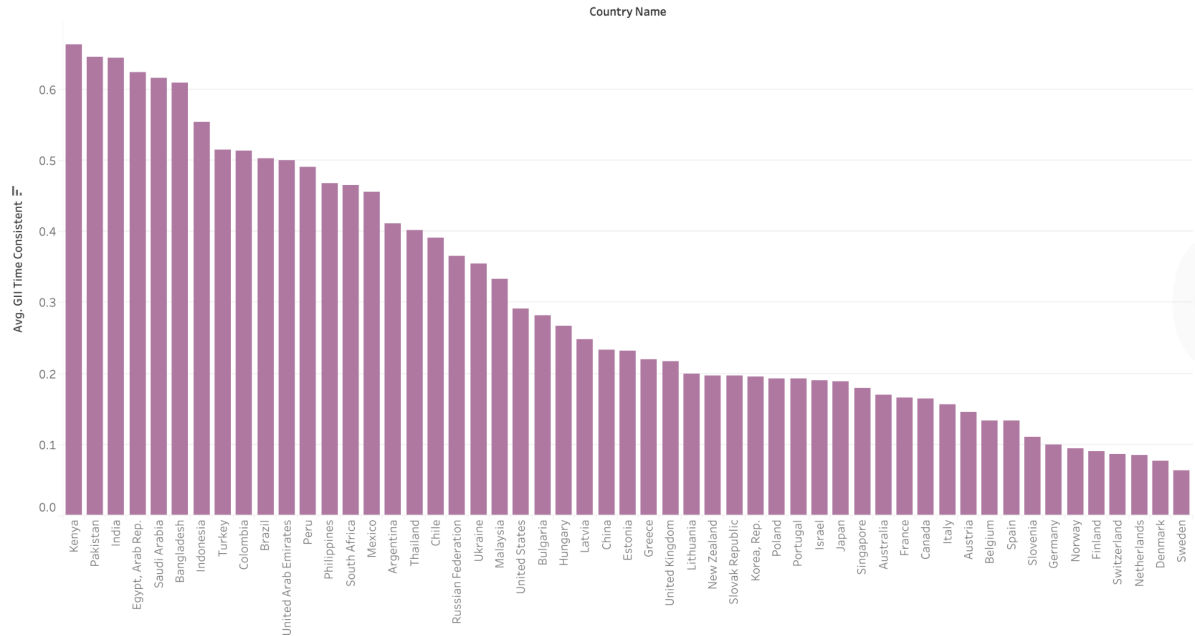


Figure 7. Gender Inequality Index by Country (Average GII, lower = more equal)

Figure 7 ranks countries by their average Gender Inequality Index (GII). Higher values indicate greater inequality, which we can see countries like Kenya, Pakistan, and India with the highest GII values (above 0.6) reflecting the severe structural barriers women face in those countries. We can suggest that it is due to factors like restrictive societal/cultural norms, limited access to education, and weak legal protection. In contrast, Sweden, Denmark, the Netherlands, and Finland display the lowest GII values (below 0.1), consistent with their status as high-innovation, high-income economies due to their inclusive legal and education frameworks. This distribution supports our analysis by showing lower inequality is consistently associated with the same countries leading in innovation/economic performance.

Female Inclusion Cont.

Creation of Opportunities vs Females in Management
(Including % of GDP on R&D)

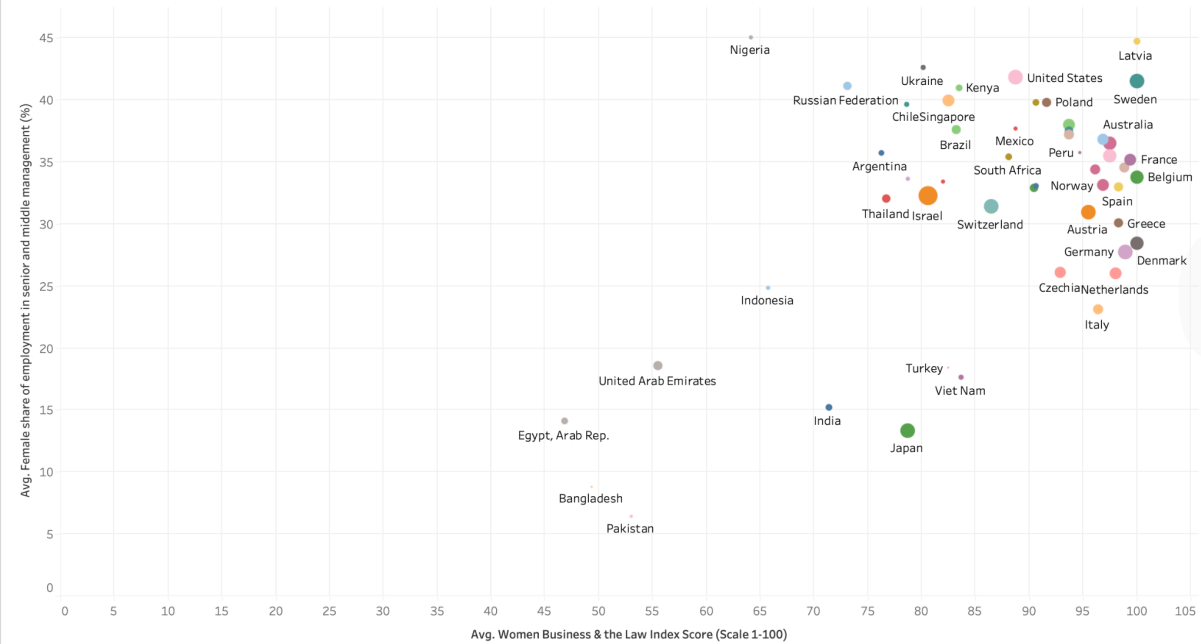


Figure 8. Creation of Opportunities vs Females in Management (Including % of GDP on R&D)

Figure 8 examines the relationship between legal frameworks and their representation in senior and middle management. We can see a general pattern where countries with stronger legal protections tend to translate to more female representation through leadership positions, but we cannot say it is a consistent pattern. Take Japan for example, they are an exception to this pattern as this country highlights the gap between formal legal equality and the actual workforce outcome. Despite that exception, we still see some consistency. For example, if we take a look at Pakistan, their weaker legal frameworks reinforces our idea of accessibility. Their lack of legal access can compound existing barriers. This indicates policies alone are not sufficient enough to combat the obstacles structural and cultural factors raise.

Female Inclusion Cont.

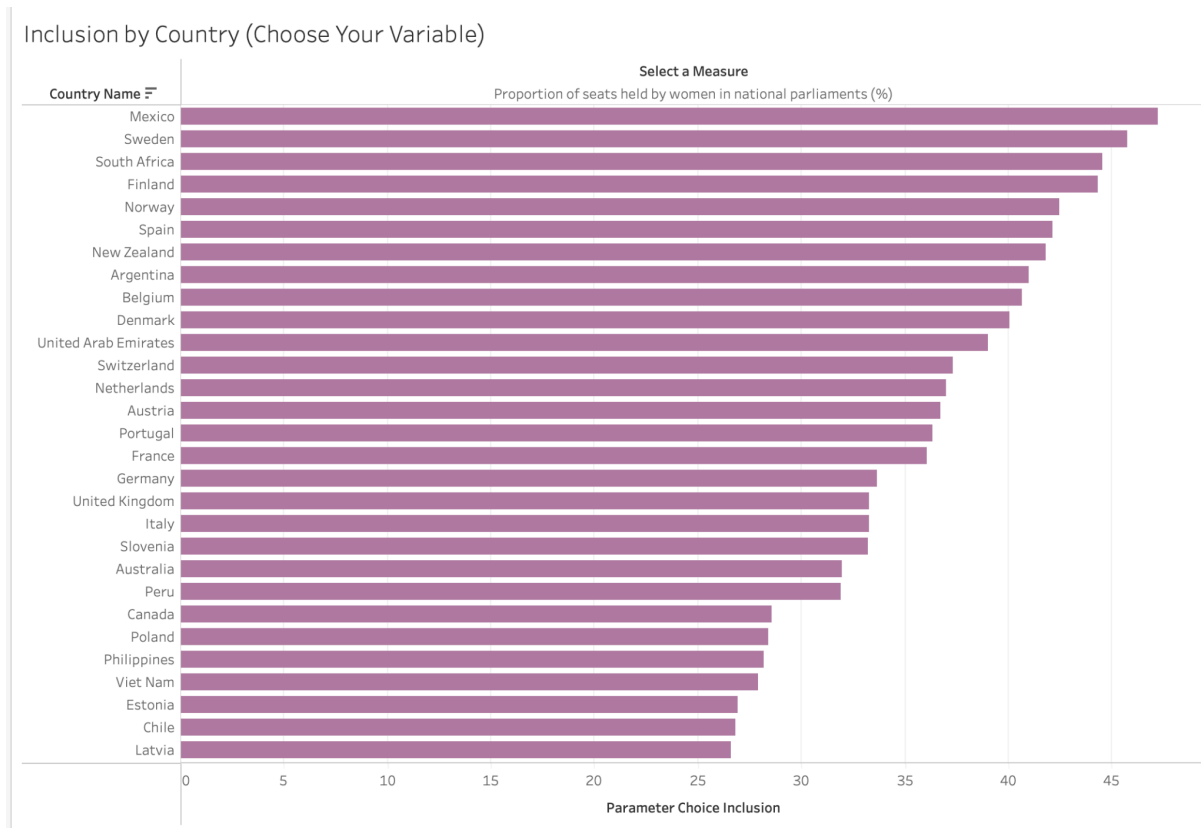


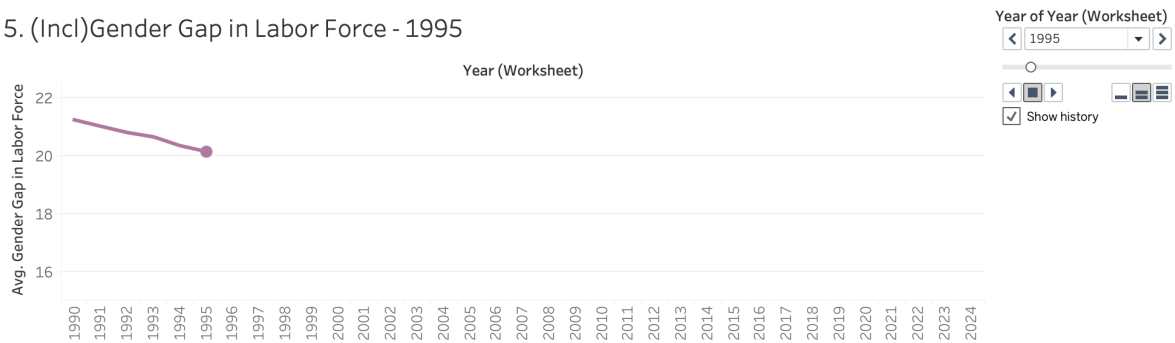
Figure 9. Inclusion by Country – Choose Your Variable (Parameter: % Seats in Parliament shown)

Figure 9 enables country-level comparison across four inclusion measures via a parameter selector: female labor force participation, Women Business and Law Index, proportion of seats held by women in national parliaments (%), and female unemployment rate.

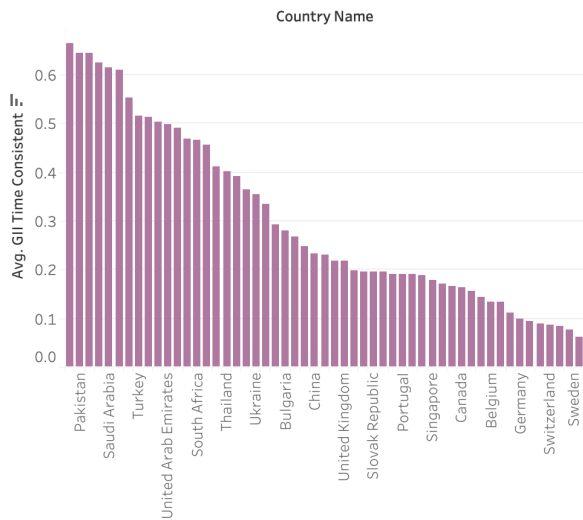
Our goal is to show that inclusion is not uniform across all countries and take note how strong performance in one area does not always translate to strength in others. We are reinforcing that women's inclusion is multidimensional and highlighting progress in one area does not guarantee equality across all systems. We can observe the progress being made, but the conclusion is that what's currently done is still not enough.

Female Inclusion Cont.

5. (Incl)Gender Gap in Labor Force - 1995



Gender Inequality Index by Country



Creation of Opportunities vs Females in Management (Including % of GDP on R&D)

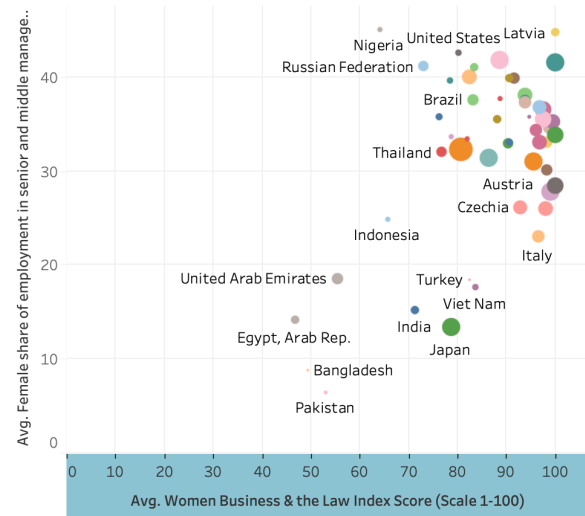


Figure 10. Inclusion Dashboard

The Inclusion Dashboard (Figure 10) shows the progression in inclusion is still uneven across countries. From this we gather that reducing inequality requires change structurally and requires policy support. Together this view shows us how inclusion is not driven by a single factor, but instead by interaction of multiple systems.

Combined Assessment

Linking Innovation and Economic Prosperity to Inclusivity

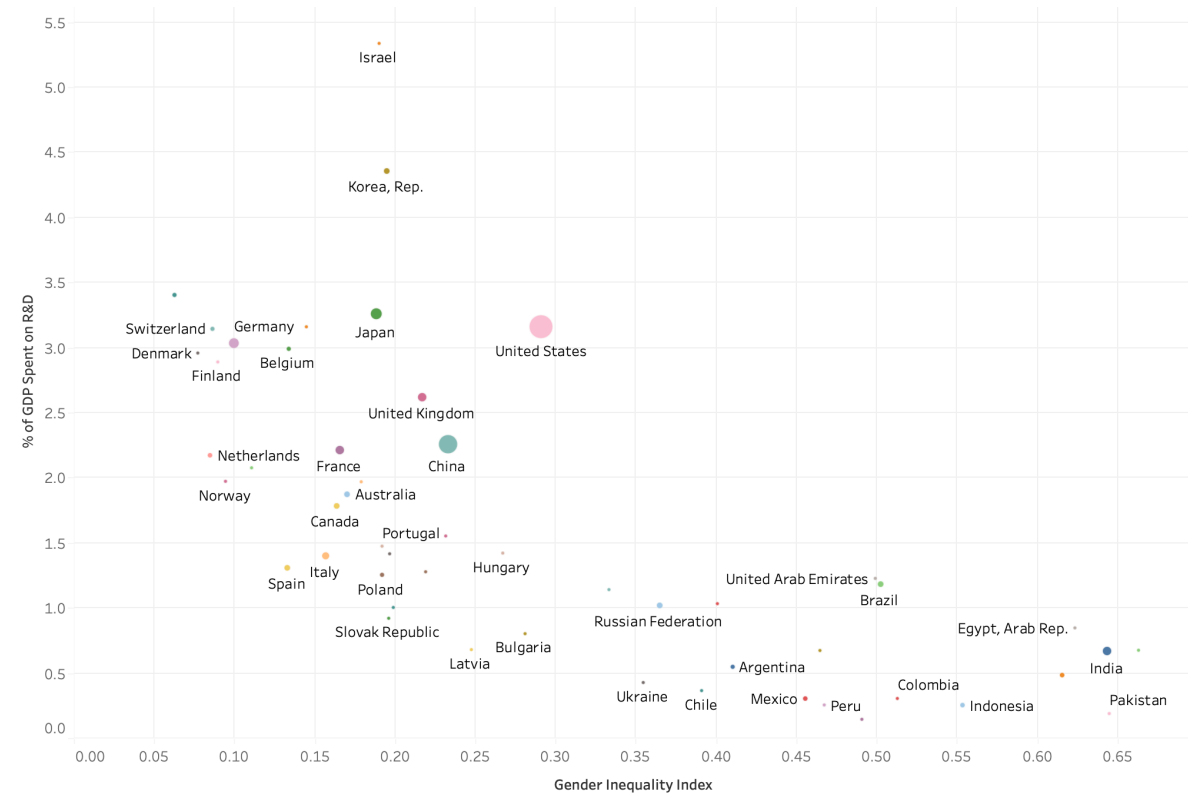


Figure 11. Linking Innovation and Economic Prosperity to Inclusivity (R&D % GDP vs Gender Inequality Index, bubble size = GDP)

Figure 11 brings together the core aspects of inequality, innovation investment, and economics to show how these factors interact across countries. We can see a clear pattern where countries with lower GII tend to invest more in R&D, which suggests more inclusive systems are associated with stronger innovation capacity.

Once again the relationship is not perfect because of the differences in a country's priorities with respect to their economic resources and scale. We know that each country has its own influences alongside their economic capacity and chosen investment strategy. For analysis purposes we can say that lower inequality is associated with higher innovation investment, but inclusion is a critical component of a broader system.

Conclusion

Even though women are increasingly gaining the qualifications needed to participate, what we found in the data highlights the role of institutional and structural factors. These factors ultimately shape whether those opportunities translate into true participation and representation.

To answer our research questions:

1. To address our main question, innovation-driven economies generally tend to show higher levels of women's inclusion, especially when investment is supported by more inclusive systems.
2. In terms of efficiency, excluding women from participation can limit efficiency by restricting the full use of available human capital.
3. Finally, we also see a positive relationship between women's inclusion and economic performance as more inclusive economies tend to align with stronger outcomes.

To conclude, we were able to reinforce that inclusion is not only for societal progress, but vital for economic performance. Innovation-driven economies are capitalistic environments that rely on human capital. It is inefficient for a country to want to generate economic growth, but have a flawed system that does not utilize the capability of all individuals.